

Introduction to Social Data Analytics:

Lecture 5 Part 2

Today's Lecture

- ▶ Logic in R
- ▶ Subsetting data
- ▶ Estimating the disparity in callback rates between white and Black applicants from Bertrand and Mullainathan (2004)

Logical operators in R

Logical operators allow you to make logical statements that produce TRUE/FALSE objects

- ▶ ! – not
- ▶ & – and
- ▶ | – or
- ▶ == – equals
- ▶ != – not equals
- ▶ > – greater than
- ▶ < – less than
- ▶ >= – greater than or equal to
- ▶ <= – less than or equal to

Logical operators with numbers in R

- ▶ `==` asks if two things are equal

```
5 == 6
```

```
## [1] FALSE
```

```
7 == 7
```

```
## [1] TRUE
```

- ▶ `!=` asks if two things are not equal

```
5 != 6
```

```
## [1] TRUE
```

```
7 != 7
```

```
## [1] FALSE
```

Logical operators with numbers in R

- ▶ `>` asks if one number is greater, `>=` greater than or equal

```
5 > 6
```

```
## [1] FALSE
```

```
7 >= 7
```

```
## [1] TRUE
```

- ▶ `<` asks if one number is less than, `<=` less than or equal

```
5 < 6
```

```
## [1] TRUE
```

```
7 <= 7
```

```
## [1] TRUE
```

Logical operators with numbers in R

- ▶ Example of & operator

```
7<8 & 6<5
```

```
## [1] FALSE
```

- ▶ Read this as “7 is less than 8 and 6 is less than 5?” Because 6 is not less than 5 this statement is FALSE
- ▶ Example of | operator

```
7<8 | 6<5
```

```
## [1] TRUE
```

- ▶ Read this as “7 is less than 8 or 6 is less than 5?” Because 7 is less than 8 this statement is TRUE

You can perform logic on vectors

- Let's go back to our graduation date example

```
graduationdate <- c(2010, 2009, 2015, 2001)
```

```
#Who graduated after 2010?
```

```
graduationdate>2010
```

```
## [1] FALSE FALSE  TRUE FALSE
```

```
#Who graduated in 2001?
```

```
graduationdate==2001
```

```
## [1] FALSE FALSE FALSE  TRUE
```

Relationship between logical and numeric objects

- ▶ When used in numeric functions, R will consider TRUE to be 1 and FALSE to be 0

#R converts TRUE into a 1

```
as.integer(TRUE)
```

```
## [1] 1
```

#and FALSE into a 0

```
as.integer(FALSE)
```

```
## [1] 0
```


Relationship between logical and numeric objects (cont)

- ▶ When used in numeric functions, R will consider TRUE to be 1 and FALSE to be 0

```
#Say we created a logical vector  
logic <- c(TRUE, TRUE, FALSE, TRUE)  
#R will calculate the mean and sum as if  
#TRUE were 1 and FALSE were 0  
mean(logic)
```

```
## [1] 0.75
```

```
sum(logic)
```

```
## [1] 3
```

Subsetting

- ▶ `[]` is for subsets
- ▶ For vectors, `[]` identifies the element we want to extract
- ▶ Inside `[]`, we can use a number for the index of the element `graduationdate[3]`
- ▶ Or we can use logic `graduationdate[school=="UCSD"]`

Subsetting with the element index

To access specific elements of a vector, we use square brackets with a number [3]. This is called *indexing*:

```
#Vector of graduation date
```

```
graduationdate <- c(2010, 2009, 2015, 2001)
```

```
#Access the third element of graduation date vector
```

```
graduationdate[3]
```

```
## [1] 2015
```

```
#Access the first element graduation date vector
```

```
graduationdate[1]
```

```
## [1] 2010
```

Accessing more than one element from a vector

We can use a vector of indices to extract multiple elements from a vector

```
#Access first and third elements for graduation date vector  
graduationdate[c(1,3)]
```

```
## [1] 2010 2015
```

Using logic to extract elements from a vector

We can use a vector of logic to extract elements of another vector

```
#Access first and third elements for graduation date vector  
graduationdate[c(TRUE, FALSE, TRUE, FALSE)]
```

```
## [1] 2010 2015
```

This allows us to extract using logical statements

```
school <- c("UCSD", "UCB", "UCLA", "UCR")  
school=="UCSD"
```

```
## [1] TRUE FALSE FALSE FALSE
```

```
graduationdate[school=="UCSD"]
```

```
## [1] 2010
```

Just like subsetting vectors, we can use logic to subset data frames

```
students <- data.frame(school=c("UCSD","UCB","UCSD"),  
                        graduationdate=c(2010,2019,2015))
```

- ▶ Now when we index, need to specify both row and column
- ▶ `student[i,j]` is the element in the i^{th} row and j^{th} column

Extract elements of our data

- ▶ Extract second row, second column

```
students[2,2]
```

```
## [1] 2019
```

- ▶ Extract third row

```
students[3,]
```

```
##   school graduationdate
```

```
## 3    UCSD           2015
```

- ▶ Extract second column

```
students[,2]
```

```
## [1] 2010 2019 2015
```

We can also use logical statements to subset data

- ▶ Extract students that attend UCSD

```
students[students$school=="UCSD",]
```

```
##   school graduationdate  
## 1   UCSD             2010  
## 3   UCSD             2015
```

- ▶ Same approach but using subset() function

```
subset(students, school=="UCSD")
```

```
##   school graduationdate  
## 1   UCSD             2010  
## 3   UCSD             2015
```

- ▶ saving a new df

```
ucsd_students <- subset(students,school=="UCSD")
```


Resume Experiment: Putting it All Together

- ▶ Now that we understand how to use logic to subset dataframes, let's go back to the empirical application of the week from Bertrand and Mullainathan (2004)
 - ▶ Do callback rates of Black and white applicants differ in an experiment in which the perceived race of the applicant is randomized?
- ▶ First step: Load the data

```
resume <- read.csv("resume.csv")
```

Summary of the dataset

```
summary(resume)
```

```
##           X           firstname           sex           race
##  Min.      :    1  Length:4870      Length:4870      Length:4870
##  1st Qu.:1218   Class :character  Class :character  Class :character
##  Median :2436   Mode  :character  Mode  :character  Mode  :character
##  Mean    :2436
##  3rd Qu.:3653
##  Max.    :4870
##           call
##  Min.      :0.00000
##  1st Qu.:0.00000
##  Median :0.00000
##  Mean    :0.08049
##  3rd Qu.:0.00000
##  Max.    :1.00000
```

Look at the Raw Data

```
head(resume)
```

```
##      X firstname      sex  race call
## 1 1      Allison female white     0
## 2 2      Kristen female white     0
## 3 3      Lakisha female black     0
## 4 4      Latonya female black     0
## 5 5        Carrie female white     0
## 6 6          Jay   male white     0
```

Average Callback Rate

- ▶ Before turning to average callback rates by race, let's look at the overall average

```
mean(resume$call)
```

```
## [1] 0.08049281
```

- ▶ So about 8 percent of applications received a call back
 - ▶ This is relatively standard in these studies, many applications receive no call back for an interview

Use logical subsetting on the data

Let's see how many resumes had stereotypically white names and how many had stereotypically Black names.

```
#Using table
```

```
table(resume$race)
```

```
##
```

```
## black white
```

```
## 2435 2435
```

```
#Using logic
```

```
sum(resume$race=="black")
```

```
## [1] 2435
```

```
sum(resume$race=="white")
```

```
## [1] 2435
```

Subset one variable with a logical statement on another variable

Say we were interested in the call back rate of resumes with stereotypically white names.

```
mean(resume$call[resume$race=="white"])
```

```
## [1] 0.09650924
```

And we want to compare to the call back rate of resumes with stereotypically black names.

```
mean(resume$call[resume$race=="black"])
```

```
## [1] 0.06447639
```

Another way to do this: create two subsetting data frames

Separate resumes with stereotypically white and black names

```
resume_blacknames <- resume[resume$race=="black",]  
resume_whitenames <- resume[resume$race=="white",]
```

Or we could do this using the subset command

```
resume_blacknames <- subset(resume, race=="black")  
resume_whitenames <- subset(resume, race=="white")
```

Calculate call back rate for each data frame.

```
mean(resume_blacknames$call)
```

```
## [1] 0.06447639
```

```
mean(resume_whitenames$call)
```

```
## [1] 0.09650924
```

- Interpretation: Black applicants about 3.2 percentage points less likely to receive a callback, despite these applications being similar **on all dimensions**

Heterogeneity

- ▶ We have found the average level of discrimination, but we can also use subsetting to explore heterogeneity by sex

```
resume_black_fem <- subset(resume, race=="black" & sex=="female")  
resume_black_male <- subset(resume, race=="black" & sex=="male")
```

```
resume_white_fem <- subset(resume, race=="white" & sex=="female")  
resume_white_male <- subset(resume, race=="white" & sex=="male")
```

Discrimination Among Female Applicants

```
# callback rate for Black and Female  
mean(resume_black_fem$call)
```

```
## [1] 0.06627784
```

```
# callback rate for White and Female  
mean(resume_white_fem$call)
```

```
## [1] 0.09892473
```

Slightly higher callback rates for female applicants, but overall level of discrimination is very similar

Discrimination Among Male Applicants

```
# callback rate for Black and Female  
mean(resume_black_male$call)
```

```
## [1] 0.0582878
```

```
# callback rate for White and Female  
mean(resume_white_male$call)
```

```
## [1] 0.08869565
```

Slightly lower callback rates for male applicants, but overall level of discrimination is very similar

Conclusion

- ▶ Recent studies have expanded on this design
- ▶ Kline, Rose and Walters (2021) sent around 83,000 applications to the top 100 U.S. Employers (this is a massive experiment)
- ▶ On average find lower callback rates for Black applicants (about 2.1 percentage points)
 - ▶ But also find significant variation across employers, with gaps concentrated in a subset of firms
- ▶ This study also discusses results with respect to gender, age, and sexual orientation